Name: ______________________

Worksheet

Volume of cylinders

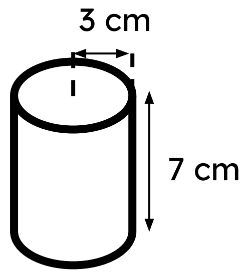
Task A: Calculating the volume of a cylinder

1) This is the formula for volume of a cylinder = $\pi r^2 h$, where r is the radius and h is the height. Which of the following are equivalent?

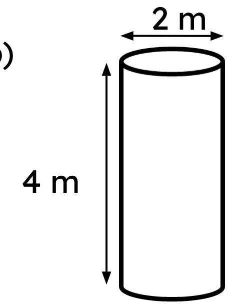
a	$V = (\pi r)^2 h$	**e**	$V = \pi \times r \times r \times h$
b	$V = \pi h r^2$	**f**	$V = \pi \times h \times r \times r$
c	$V = \pi r h^2$	**g**	$V = \pi^2 r h$
d	$V = \pi (rh)^2$	**h**	$V = r^2 h \pi$

2) Calculate the volume of the following cylinders. Leave in terms of π.

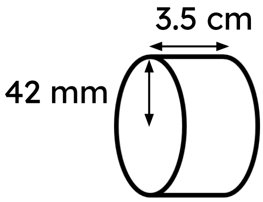
a)



b)



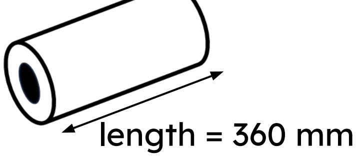
c)



d) A cylinder with a radius of 8 mm and a depth of 12 mm

e) A cylinder with a diameter of 4.8 cm and a height of 12 cm

3) A metal washer factory forms a long metal pipe before slicing them into individual washers. What volume of metal is used to create this section of pipe?



internal diameter = 12 mm
external diameter = 35 mm

Task B: Using the volume of a cylinder

1) All of these cylinders have the same volume. Calculate the missing lengths.